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INFORMATIKA ANGOL NYELVEN

EMELT SZINTŰ GYAKORLATI VIZSGA

minden vizsgázó számára

2022. május 16. 8:00

Időtartam: 300 perc

Beadott dokumentumok	
Piszkozati pótlapok száma	
Beadott fájlok száma	

A beadott fájlok neve

EMBERI ERŐFORRÁSOK MINISZTERIUMA

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Important information

Devices allowed for the exam: computer assigned to the candidate, paper, pen, pencil, ruler, sealed notepaper.

You can take **notes** on the internal pages of the exercise sheet and the notepaper, these should be submitted at the end of the exam but their content will not be evaluated.

The exercises **can be solved in any order**.

Please pay attention to **frequent saving** (every 10 minutes); it is suggested that you save your work every time you start a new exercise.

Save your exam work **in the exam directory that corresponds to the identifier** given on your exercise sheet. Check that the directory that corresponds to the code is accessible; if it is not accessible, notify the supervising teacher at the beginning of the exam.

Save your works **in the exam directory** and at the end of the exam **check** that every solution is in the given directory because only those solutions can be evaluated. Check that the files to be submitted are readable because files that cannot be opened cannot be evaluated.

If you solve the database management exercise with LibreOffice Base, then you should submit the SQL commands describing update queries either as part of the LibreOffice Base database file or as a separate text file. If you submit them as a text file, the name of the text file should refer to its contents clearly (e.g. *SQL_commands.txt*) and the required query name should be displayed next to the command in the file.

If you use a **MySQL** database engine, the data of the database should also be saved in a so-called **“dump” file**.

The submitted program can be evaluated only if the candidate created the source file(s) that correspond(s) to the chosen programming environment in the exam directory and it/they contain(s) the source code that belongs to the solution of the exercise parts.

The **source files** can be found in the exam directory.

It is suggested that you **read through** the exercises first, then solve the individual exercise parts one by one.

If your computer has **technical problems**, indicate it to the supervising teacher. The fact of indication and the observed problem will be recorded. The lost time will be added to the duration of the exam. If the problem is not of computer nature, the correcting teacher should take the description of the case in the record into consideration. (The system administrator cannot help the candidate with the solution of the exercises.)

At the end of the exam on the first page of the exam document you should indicate **the number and name of files created and submitted by you and located in the exam directory and its subdirectories**. When finishing the exam, do not leave the room until you have done so and have shown it to the supervising teacher.

Please indicate the operating system and the programming environment you work with:

Operating system: ☐ Windows ☐ Linux

Programming environment:

- | | | |
|----------------------------------|------------------------------|-------------------------------------|
| ☐ FreePascal | ☐ GCC | ☐ Visual Studio |
| ☐ Lazarus | ☐ Perl 5 | ☐ _____ |
| ☐ JAVA SE | ☐ Python | ☐ _____ |

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1. Ervin Lázár

Ervin Lázár is a Kossuth Prize winner Hungarian writer, storyteller, fabulist, founding member of the Digital Literary Academy. The Petőfi Literary Museum has a wide collection of his works and documents about him. Create two web pages using these materials according to the example and the description, which give a brief introduction of the writer.

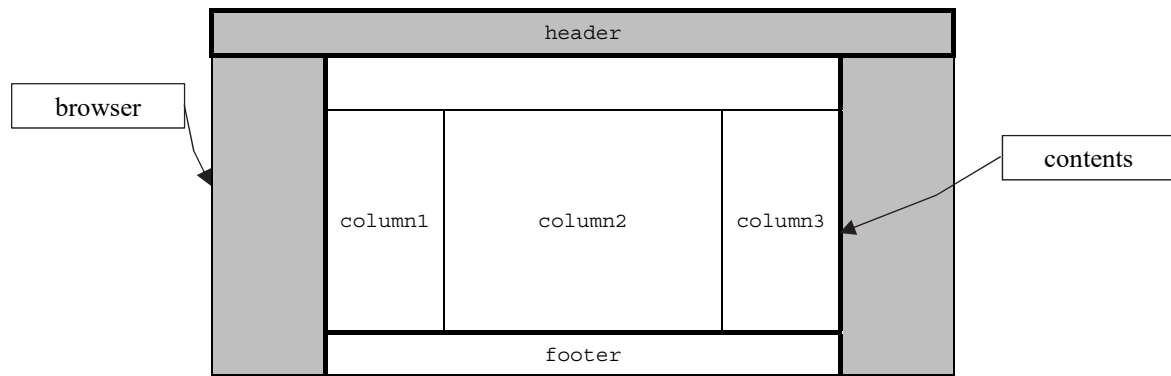
- The names of the files to be created are *ervinlazar.html* and *biography.html*.
- The source of file *biography.html* can be found in file *le_source.html*. A part of file *le_source.html* should also be used for the creation of file *ervinlazar.html*.
- The pictures required for the solution of the exercise are *dia.png*, *le_picture.jpg* and *szegenydzsoni.jpg*. When using the pictures, pay attention to the pictures being displayed correctly even if the exam directory is moved.

You can create the web pages with style sheets or without them. If you use a style sheet, then you can use file *le_style.css* to format both web pages, but you have to make further settings and amendments in the file.

1. Create the following uniform colour structure on both pages. The background colour is white, the font colour of the text and the links in every state is dark grey (colour code #0E0E0E), the font colour of the titles and the subtitles is red (#EB6160).
2. The font type of both web pages is Verdana, the paragraphs are justified.
3. First create page *biography.html*. Use file *le_source.html*, which contains the text in a partly formatted form. Display text “Biography” in the header of the browser.
4. The contents of the page should be displayed as a box of width 1000 pixels without border, aligned centred. This should be achieved by using a table or by setting and applying the *contents* identifier in the style sheet. The background should be white and this setting should appear in the code of the web page or the style sheet.
5. The Heading-1 style of the titles and the Heading-2 style of the subtitles are already set in the source. Complete the formatting so that the title and the three subtitles are displayed with the red colour of the given code. The font size of the subtitles should be 17 pixels. Set that the titles are separated by a distance of 5 pixels from every other part of the page in all four directions.
6. Create an identifier or reference with name “*awards*” for subtitle “*Major awards, recognitions:*”, which will be required for the reference to be created on page *ervinlazar.html*. Convert the reference at the bottom of the page into a link that leads to the given address.
7. Then create page *ervinlazar.html*. The figure shows the structure of the browser window. The header of the page should fill the browser window horizontally. The contents should be created by using a table or by setting and applying styles *header*, *contents*, *column1*, *column2*, *column3* and *footer* in the style sheet according to the followings:
 - the box (the white area marked as contents in the figure) should be white, without border and 1000 pixels wide,
 - the box is aligned centred horizontally, the contents are displayed at the top of it,
 - the width of the first and third columns is 220 pixels, and the width of the second is 560 pixels.

The exercise continues on the next page.

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8. The source text of the page can be found laid out into paragraphs and partly formatted in the code of file *biography.html*, or unformatted in file *le_source.html*, insert the appropriate parts into the file according to the example. Insert the text into part *column1* from the comments in the source file. The title of the web page to be displayed in the header of the browser is “Ervin Lázár”. The formatting of the title and the subtitles should be the same in the two files.
9. Format the header or set its properties using the class identifier *header* in the style sheet:
 - the text is aligned right and has an indent of 10 pixels on all sides,
 - the background is the given red colour and the text colour is white,
 - the font size is 30 pixels and the font style is bold.
10. In the following exercises format all three inserted pictures or use the class identifier *picture* in the style sheet to set their properties uniformly according to the followings:
 - the width is 190 pixels keeping the aspect ratio,
 - no border,
 - a distance of 10 pixels on the left and the right.
11. In the beginning of part *contents*, above the columns, insert the logo of the Digital Literature Academy, file *dia.png* and insert text “Member of the Digital Literature Academy”, whose style is Heading-1 and font colour is the given red colour.
12. Insert the photo of Ervin Lázár, *le_picture.jpg* into part *column1*. Below the photo, convert texts “**Biography**” and “**Major awards**” into links. The first should lead to *biography.html* and the second should lead to identifier “*awards*” in it.
13. Part *oszlop2* should contain some of the biography and of the major awards. The additional information should be indicated by setting the font colour of the part shown in the example to light grey (colour code #C7C7C7), or by using class identifier *grey* in the style sheet.
14. Insert text “More” into the two positions shown in the example and convert them into links that lead to page *biography.html*.
15. Insert picture *szegenydzsoni.jpg* into the part *column3*. Create the caption with the layout shown in the example under the picture, its text is “Poor Johnny and Arnika (1981)”. Format it or use class identifier *caption* in the style sheet. Set the font size to 9 pixels, the wrapping into three lines and the centred horizontal alignment.
16. Insert a horizontal line of length 50% aligned centred in the footer, above subtitle “**Source:**”.
17. Convert the reference into a link that leads to the given address.

30 marks

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Example for the Ervin Lázár exercise:

Ervin Lázár (1936–2006)

Biography

He was born on 5 May 1936 in Budapest. His father was István Lázár, a manorial administrator, and his mother was Etelka Pentz. He spent his childhood at Alsó-Ráceszpuszta in Tolna County, where the Lázár family lived until 1951. He went to school in the surrounding villages. From October 1950 he continued his studies at the János Garay High School in Szekszárd and graduated there in 1954. He began higher education in his graduation year at the Faculty of Humanities of Eötvös Loránd University in Budapest, majoring in journalism. He lived in the Eötvös College. In 1956 he completed his compulsory internship for third-year students in Nyíregyháza, then in 1957 he moved to Pécs.

As he was employed as a journalist by Esti Pécsi Napló from 1 February 1959, in December of the same year, he took a differential exam and continued his studies at the Hungarian major as a correspondent student. He graduated as a teacher of Hungarian in 1961.

From 1959 to 1963 he worked for Esti Pécsi Napló. Between 1 January 1964 and 15 March 1965 he was employed by Dunántúli Napló and by Jelenkor at the same time. As a journalist, his real genre was the report. In March 1965 he moved to Budapest: he joined Élet és Irodalom as a layout editor, where he was employed until 30 April 1971. From then until 1989 he was a freelance writer. In the meantime, he and his family lived in Pécel for nine years, before moving permanently back to the capital in 1980.

In 1989, until the publication of the first issue of the paper, he was a member of the editorial board of Új Idők. A founding member of Magyar Fórum; he worked for the paper from 1 October 1989 to 10 August 1990 as a full-time employee. Then, from 11 August 1990 to 9 May 1991, he worked as a reading editor for Magyar Napló, from 10 May 1991 to 15 January 1992, for Pesti Hírlap and from 16 January 1992 to 31 August 1992 for Magyar Nemzet. From September 1992, he was the reading editor of Hítel. He retired at the age of sixty.

He was a member of the National Association of Hungarian Journalists from 1959 to 1994, where he also participated in the work of the ethics committee between 1991 and 1994. He joined the Hungarian Writers' Association in 1969 and was a member of the electorate for a long time. He was a member of the Hungarian Academy of Arts from 1994.

For him, the fundamental experience of the childhood was decisive: he admittedly drew most of his inspirations from there. His style is cinematic; usually, his texts almost offer themselves as a basis for radio, stage and film.

biography.html

Ervin Lázár (1936–2006)



Member of the Digital Literature Academy



Biography

He was born on 5 May 1936 in Budapest. His father was István Lázár, a manorial administrator, and his mother was Etelka Pentz. He spent his childhood at Alsó-Ráceszpuszta in Tolna County, where the Lázár family lived until 1951. He went to school in the surrounding villages. From October 1950 he continued his studies at the János Garay High School in Szekszárd and graduated there in 1954. He began higher education in his graduation year at the Faculty of Humanities of Eötvös Loránd University in Budapest, majoring in journalism. He lived in the Eötvös College. In 1956 he completed his compulsory internship for third-year students in Nyíregyháza, then in 1957 he moved to Pécs.

As he was employed as a journalist by Esti Pécsi Napló from 1 February 1959, in December of the same year, he took a differential exam and continued his studies at the Hungarian major as a correspondent student. He graduated as a teacher of Hungarian in 1961.

[More](#)

Major awards, recognitions:

- 1974 – Attila József Prize
- 1980 – Literary Prize of the Arts Fund
- 1981 – State Youth Award
- 1982 – Andersen Diploma
- 1985, 1989, 1996 – Book of the Year Award

[More](#)

Biography

Bibliography

Subject literature

Digitized works

Paintings

Photos

Covers

Manuscripts

Major awards

Recommended pages

Source:

<https://pim.hu/hu/dia/hu/dia-tagjai/lazar-ervin>



Poor Johnny
and Arika
(1936)

ervinlazar.html

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2. Accommodation

In this exercise you have to process the bookings of a location specialising in accommodating class trips using a spreadsheet processor. File *accommodation.txt* presently contains the data of 9 bookings. The location accepts at most 20 classes in a year, and at a given time at most one class can be at the location. You should create the table so that the formulas return the correct result even if the data of a new group are added.

The explanation of the first row of the example in the order of the fields: the identifier of the group, where the group comes from, when they arrive and when they depart. The next three data give how many times the group request different meals at the location. The last two data give the number of boys and girls in the class, which is important for the assignment of rooms.

id	from	arrival	departure	breakfast	lunch	dinner	boys	girls
1	Budapest	2020.05.14	2020.05.16	2	2	2	14	20

Next to the data of the group, the source contains the price of the accommodation and the price of the different meals per person.

During the solution take the followings into consideration:

- *You can perform auxiliary calculations to the right of column Q. Whenever possible, use a formula, function or reference in the solution to get the correct result even if the base data change.*
 - *There are parts in the exercise that use the results from a previous question. If you could not solve the previous part completely, use its solution as it is, or enter a reasonable result and work on with that value. This way you can receive marks for that exercise part as well.*
1. Import text file *accommodation.txt*, which is UTF-8 encoded and tagged by tabs, into the worksheet of the spreadsheet processor starting from cell *A1*. Save your work in the default format of the spreadsheet processor as *accommodation*.
 2. In the cells of range *J2:J21*, determine the total number of students in each group.
 3. In the cells of range *K2:K21*, determine the total price of the accommodation for the student groups. The formula should refer to the unit price given in cell *P1*, which is given per person per night. Create a formula that can be copied flawlessly.
 4. In the cells of range *L2:L21*, determine the total price of the meals for the student groups. The formula should refer to the unit prices of the meals and you should be able to copy it flawlessly.
 5. In the cells of range *M2:M21*, determine the number of rooms required by the students of the group. One room can accommodate 4 students and take into account that boys and girls should not be mixed in a room.
 6. To check the booking of a new group, in cell *P9* enter the day of the arrival of the group and in cell *P10* the day of the departure. In cell *P11*, create a formula that displays word “yes” if the planned period does not overlap with any of the groups already in the table. If the period overlaps with any of the groups already in the table, the formula should display the word “overlap”. The formula should be correct for the maximum possible number of groups, which is 20. Take into account that the departure of a group and the arrival of the next group may be on the same day.

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7. Format the table according to the description and the example.

- Use thin border in the range *A1:M21*. Use double-line border under the cells of ranges *A1:M1* and *O10:P10*.
- Set currency format without decimal digits in the cells of ranges *K2:L21* and *P1:P4*.

8. In the cells of range *A2:M21*, each data of the groups whose total number exceeds the limit given in cell *P6* should be highlighted by yellow background. The background colour should remain correct even if the limit is changed.

15 marks

Example:

	A	B	C	D	E	F	G	H	I
1	id	from	arrival	departure	breakfast	lunch	dinner	boys	girls
2	1	Budapest	2020.05.14	2020.05.16	2	2	2	14	20
3	2	Pécs	2020.06.01	2020.06.03	2	1	1	13	12
4	3	Budapest	2020.06.08	2020.06.09	1	0	1	16	13
5	4	Eger	2020.06.09	2020.06.12	3	2	2	25	9
6	5	Sopron	2020.06.13	2020.06.15	2	0	1	12	12
7	6	Eger	2020.09.03	2020.09.05	2	1	2	15	11
8	7	Győr	2020.09.05	2020.09.06	1	0	1	12	11
9	8	Vác	2020.09.06	2020.09.08	2	2	2	11	13
10	9	Budapest	2020.09.10	2020.09.13	3	2	3	10	15
11	10								
12	11								
13	12								
14	13								

H	I	J	K	L	M	N	O	P
boys	girls	total	accommodation	meals	number of rooms		accommodation	3 000 Ft
14	20	34	204 000 Ft	217 600 Ft	9		breakfast	700 Ft
13	12	25	150 000 Ft	97 500 Ft	7		lunch	1 400 Ft
16	12	28	84 000 Ft	50 400 Ft	7		dinner	1 100 Ft
25	8	33	297 000 Ft	234 300 Ft	9			
12	12	24	144 000 Ft	60 000 Ft	6		limit	30
15	17	32	192 000 Ft	160 000 Ft	9			
12	17	29	87 000 Ft	52 200 Ft	8		New group	
11	13	24	144 000 Ft	153 600 Ft	7		arrival	2020.09.05
10	15	25	225 000 Ft	205 000 Ft	7		departure	2020.09.09
		0	0 Ft	0 Ft	0		can they come?	overlap
		0	0 Ft	0 Ft	0			
		0	0 Ft	0 Ft	0			
		0	0 Ft	0 Ft	0			

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3. Oscar-winning films

Oscars, officially called Academy Awards, are among the most prestigious awards in the film industry. It is awarded in many categories by the Academy of Motion Picture Arts and Sciences, for example the best director, best actor, best actress. The Oscar can be awarded for a film that was produced in the previous year. After a long selection procedure a few films are nominated for the prize and the jury selects the best one from these. This database contains the nominated films.

1. Create a new database named *oscar*. Import the three text files provided (*films.txt*, *connections.txt*, *makers.txt*), which are UTF-8 encoded and tagged by tabs, into the database with table names that correspond to the file names (***films***, ***connections***, ***makers***). The first line of the files contains the field names. Upon creation set the appropriate types and the primary keys.

Tables:

films (*id*, *nyear*, *won*, *hungarian*, *title*, *premiere*)

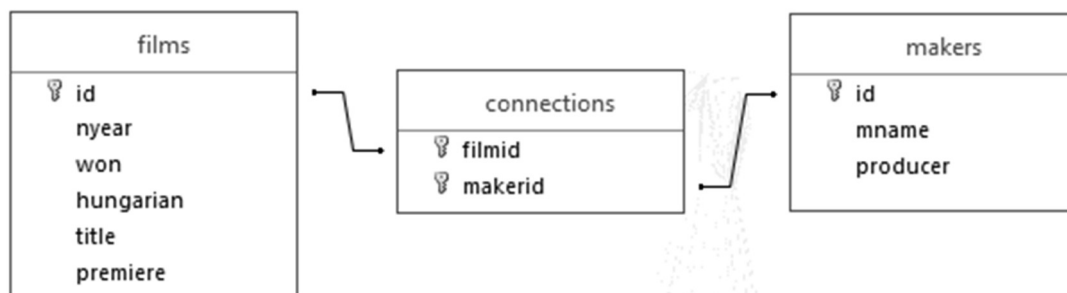
<i>id</i>	The identifier of the film (number), this is the key
<i>nyear</i>	The year of nomination for the Oscar (number)
<i>won</i>	Did the film win the Oscar (Boolean)
<i>hungarian</i>	The Hungarian title of the film, if it is known (text). It is missing in some cases.
<i>title</i>	The original title of the film (text). Titles are unique.
<i>premiere</i>	The date of the Hungarian premiere of the film, if it is known (date). In many cases it is missing.

connections (*filmid*, *makerid*)

<i>filmid</i>	The identifier of the film (number), it is a key
<i>makerid</i>	The identifier of the maker (number), it is a key

makers (*id*, *mname*, *producer*)

<i>id</i>	The identifier of the maker (number), it is the key
<i>mname</i>	The name of the maker (text). In the beginning, filming companies received the Oscar, later the producers. In the solution you can use the fact that filming companies and producers are not mixed for any of the films and that each name is unique. Only such companies and producers are included that have an associated nominated film.
<i>producer</i>	Does the name belong to a producer (Boolean)? If it belongs to a producer, then its value is true, if it belongs to a filming company, then its value is false.



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When solving the following exercises, save the queries with the name given in brackets. Pay attention to displaying exactly the required fields in the solution, do not display unnecessary fields.

2. Create a query that lists films that won the Oscar in the order of the year awarded. Display the year of award and the original title of the film. (**2winners**)
3. Create a query that determines the years when at least 10 films were nominated for the Oscar. Display only the year of nomination. (**3min10**)
4. Create a query that determines the original title of the films that were nominated in the years of World War 2 (between 1939 and 1945, including the limits) and were also premiered in this period in Hungary. (**4ww2**)
5. Create a query that determines which Academy Award-winning films were premiered at least 10 years later in Hungary than the year of winning. Display only the original title of the film. (**5later**)
6. There are producers whose several films have been nominated for the Oscar, in this exercise you have to determine these producers. Create a query that determines the period between the first and the last nomination. Display the name of the producer, the number of nominations and the period between the first and the last nomination. (**6producers**)
7. In many cases a film has several, even three or four producers. Create a query that determines "Clint Eastwood" 's co-producers for the nominated films, that is, the films that are in the database. Display the names of the people in concern and ensure that everybody is displayed only once. (**7Eastwood**)
8. Create a query that determines the names of the producers for whom the date of Hungarian premiere is not known for any of their films nominated for the Oscar. (**8notknown**)

30 marks

Sources:

1. Ervin Lázár

<https://pim.hu/hu/dia/dia-tagjai/lazar-ervin> Last access: 2 November 2020

https://pim.hu/sites/default/files/styles/uncropped_m/public/lazar_ervin.jpg?itok=PzFHRkNG Last access: 2 November 2020

3. Oscar-winning films

https://hu.wikipedia.org/wiki/Oscar-dij_a_legjobb_filmnek Last access: 29.11.2020

The pages of the different producers in database <https://www.imdb.com/> Last access: 14.12.2020

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4. Section speed control

Section speed control is used in many countries to control traffic. It is based on the vehicles passing two control gates that are several kilometres from each other. At both points the license plate number of the vehicle and the time of passing the gate are recorded. The average speed is calculated from the two time values. If it exceeds the maximum allowed speed for the given section, then the driver has violated the traffic rules. The section speed control measurement data of one of the lanes of a road section of length 10 km, which is single-lane in both directions, are available. The road section is closed, that is, vehicles cannot enter or exit it at any other point than the endpoint.

Text file *measurements.txt* contains the measurement data of a 1-hour measurement starting at 8 a.m. on a given day. Each vehicle appears only once among the measurement data. The measurement data of a given vehicle are given in one line separated by single spaces. A line contains 9 pieces of data, the license plate number of the vehicle (6 characters) and the time recorded at the entry and exit points of the section in hour, minute, second, millisecond format. (The file does not contain the data of vehicle that did not have a recorded time at the entry or the exit point of the section.)

The number of lines is at most 1000. The data are ordered by the time recorded at the entry point. For example:

```
OXZ648 8 4 44 861 8 11 53 432
QUT385 8 4 53 376 8 9 28 185
QTS988 8 5 0 854 8 12 19 879
OTP604 8 5 2 263 8 12 21 288
```

According to the example, the vehicle with license plate number QTS988 passed the recorder at the entry point at 8:5:0.854 and passed the recorder at the exit point at 8:12:19.879. Its average speed is 82 km/h, which is the distance travelled (10 km) divided by the elapsed time (0.1219 hour).

Create a program that uses the data given in file *measurements.txt* to answer the following questions. Save the source code of the program as *speedcontrol*. (When writing the program, you do not have to check the correctness and the validity of the data given by the user, you can assume that the available data correspond to the description above.)

If an exercise requires displaying data on the screen, display the exercise number before displaying the result (for example: **Exercise 3.**). If you request data from the user, display the nature of the requested data on the screen. Displaying without accents is also acceptable. When displaying decimal digits, use the default display method of the programming language used.

Display the results and communicate with the user according to the example given at the end of the exercise.

1. Read and store the data of file *measurements.txt*.
2. Display the number of vehicles whose data were recorded in the measurement on the screen.
3. From the available data, determine the number of vehicles that passed the exit point of the section before 9 o'clock. Display the number on the screen.
4. Request a time given in hour minute form from the user.
 - a. Determine the number of vehicles that passed the entry point in the given minute. If in the given minute no vehicle passed the entry point, then display value 0.
 - b. Determine the traffic intensity, which is the number of vehicle on the road section in the minute that starts at the given minute (e.g. if the given hour minute

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is 08:09, then between 08:09:00.000-08:09:59.999) divided by the length of the road section. Display the value in decimal fraction format.

- Find the speed of the vehicle with the highest average speed and the number of vehicles overtaken by it in the measured section. If there are several highest average speeds, it is enough to display only one of them. Display the license plate number of the vehicle, the average speed as an integer and the number of overtaken vehicles.
- Determine what percent of the vehicles exceeded the maximum speed limit on the measured section (90 km/h) with their average speed. Display the result in decimal number format according to the example.
- Create a text file with name *finest.txt*, which contains the data of the speeding vehicles. This text file should contain the data of vehicles whose average speed exceeds 104 km/h. The file should contain the license plate number of the vehicle, the average speed given as an integer and the value of the fine with units, separated by semicolons or tabs. The speeding fines are given in the following table.

Average speed	Fine
above 104 km/h, up to 121 km/h	30 000 Ft
above 121 km/h, up to 136 km/h	45 000 Ft
above 136 km/h, up to 151 km/h	60 000 Ft
above 151 km/h	200 000 Ft

45 marks

Example for the textual output:

```
Exercise 2.
The data of 687 vehicles were recorded in the measurement.

Exercise 3.
Before 9 o'clock 603 vehicles passed the exit point recorder.

Exercise 4.
Enter an hour and minute value: 8 20
    a. The number of vehicle that passed the entry point recorder: 12
    b. The traffic intensity: 9,4

Exercise 5.
The data of the vehicle with the highest speed are
    license plate number: OKL564
    average speed: 137 km/h
    number of overtaken vehicles: 33

Exercise 6.
11,94% percent of the vehicles were speeding.

The file is ready.
```

Contents of file *finest.txt*:

```
QZV314 128 km/h 45000 Ft
OJW811 107 km/h 30000 Ft
QUT385 131 km/h 45000 Ft
QRC440 128 km/h 45000 Ft
QJQ854 109 km/h 30000 Ft
...
```

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	mark	
	maximum	achieved
Word processing, presentation, graphics, web page composition 1. Ervin Lázár	30	
Spreadsheet processing 2. Accommodation	15	
Database management 3. Oscar-winning films	30	
Algorithmisation, data modelling 4. Section speed control	45	
Mark of the practical exam part	120	

date

examiner

	pontszáma egész számra kerekítve	
	elért	programba beírt
Szövegszerkesztés, prezentáció, grafika, weblapkészítés		
Táblázatkezelés		
Adatbázis-kezelés		
Algoritmizálás, adatmodellezés		

dátum

dátum

javító tanár

jegyző